



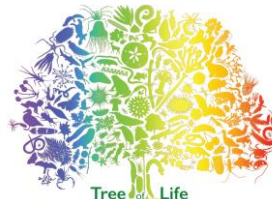
Welcome to the Biodiversity Genomics course!

Joana Meier, Karin Näsvall, Nicol Rueda
Tree of Life Programme, Wellcome Sanger Institute

Melisa Olave
CONICET, National University of Cuyo

THE
ROYAL
SOCIETY

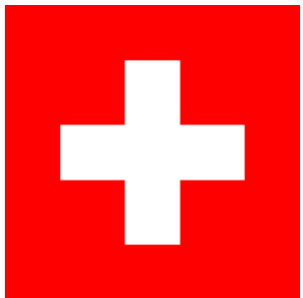
The
Branco Weiss
Fellowship
Society in Science



UNIVERSITY OF
CAMBRIDGE

Joana Meier

Childhood



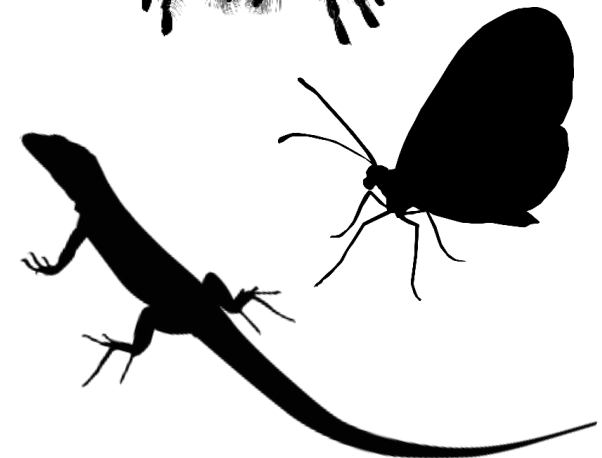
**PhD on cichlid fish
speciation in Bern,
Switzerland**



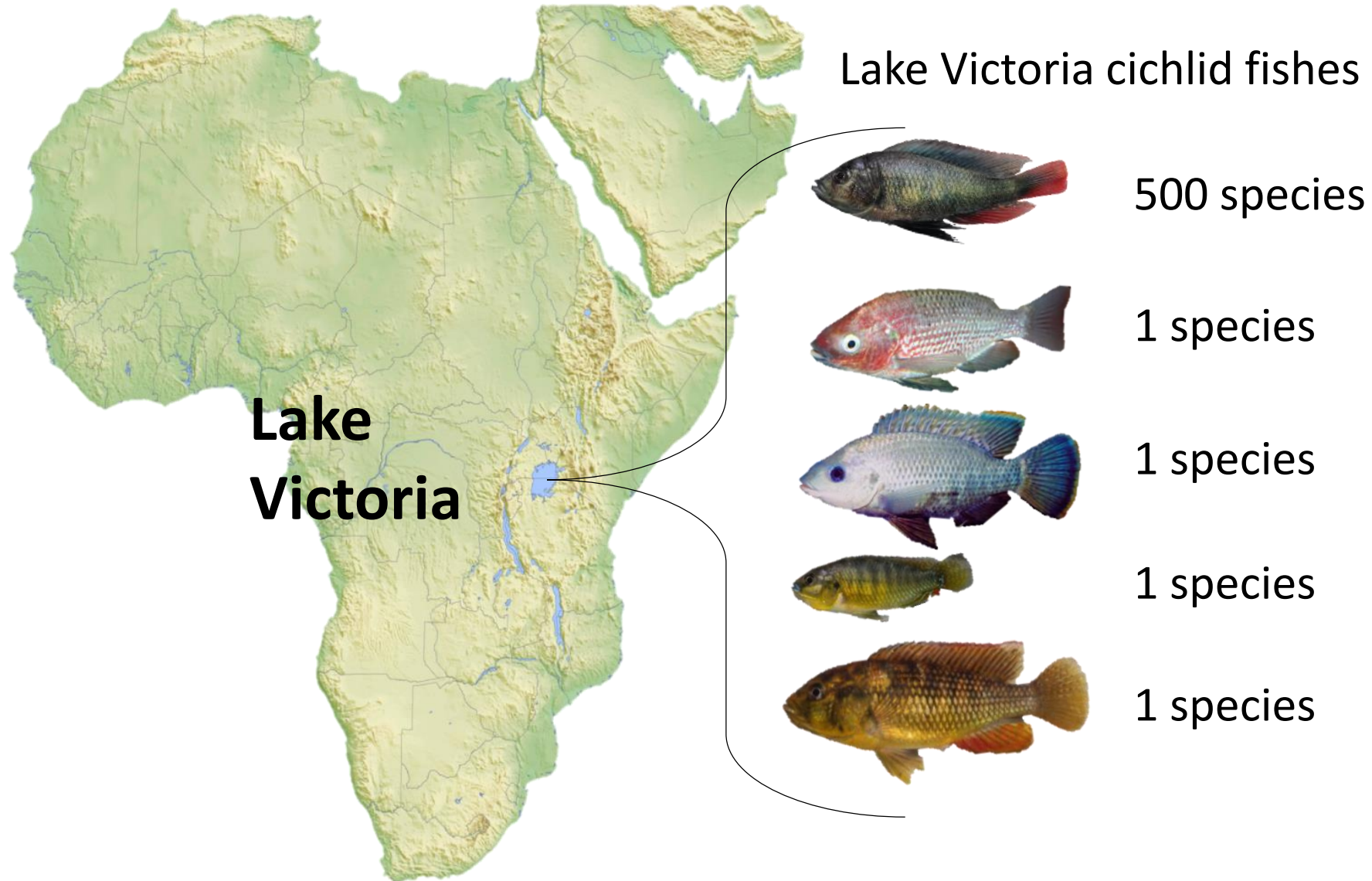
**Postdoc at the
University of
Cambridge, UK**



**Group leader at the
Wellcome Sanger
Institute**



Why do some lineages diversify very fast?



Nicol Rueda

Postdoctoral fellow

Rapid speciation group, Tree of life

Wellcome Sanger Institute

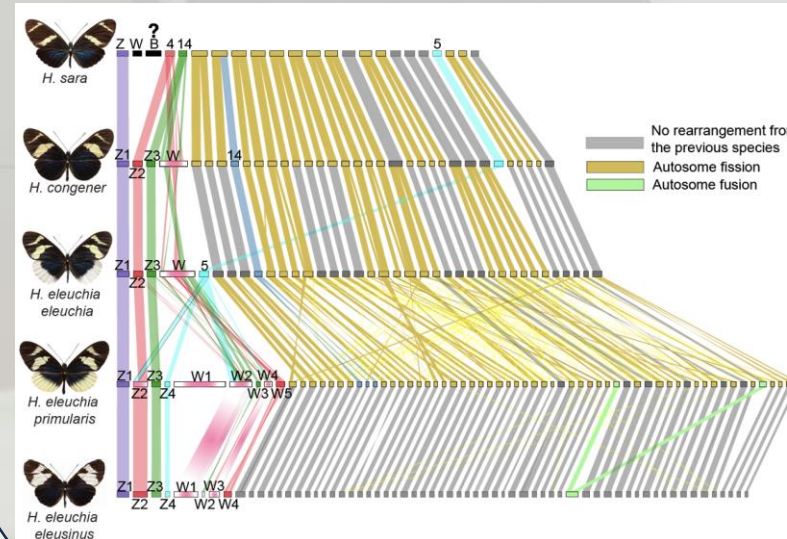
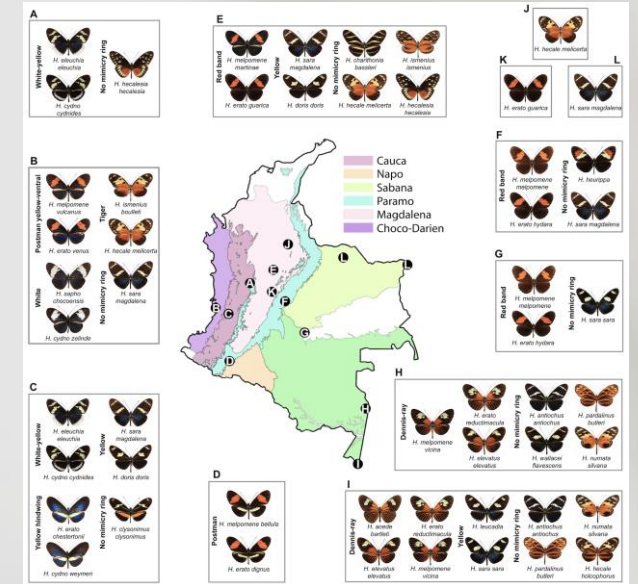
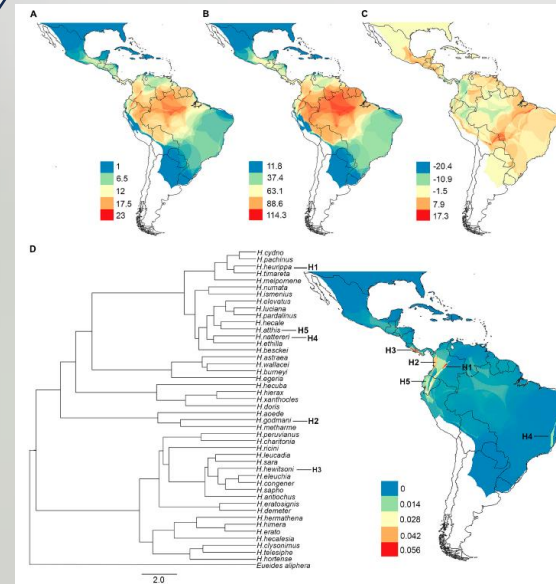
Master of science in Biology in Col/Brazil



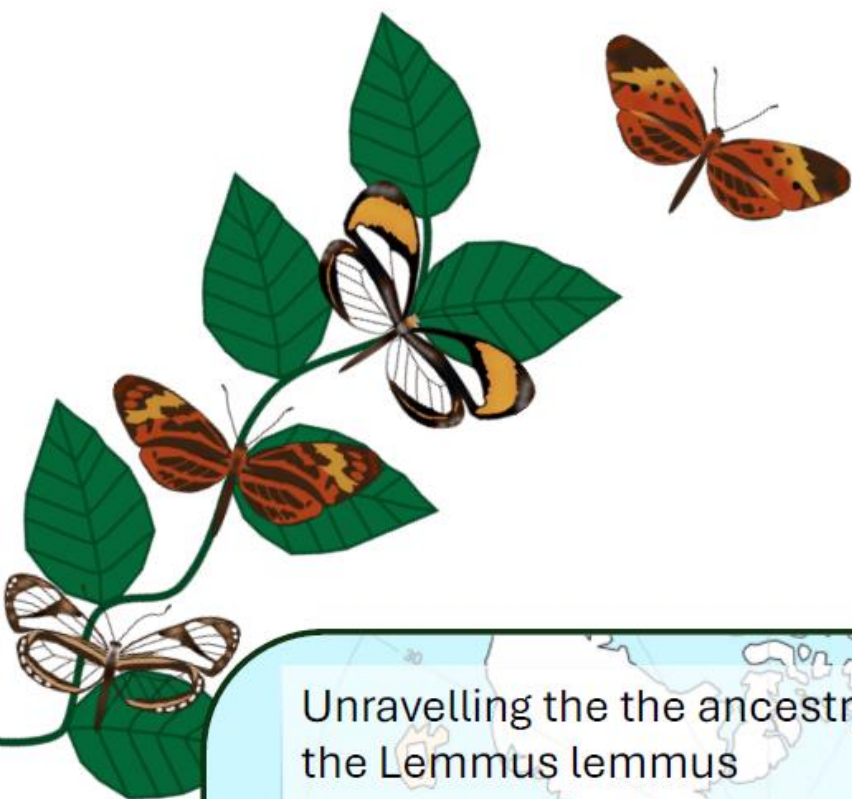
Population monitoring for 7 months in two forests of different conservation status.

Capture-Mark-recapture technique

PhD in Biology in Colombia



Sex chromosome evolution in Lepidoptera



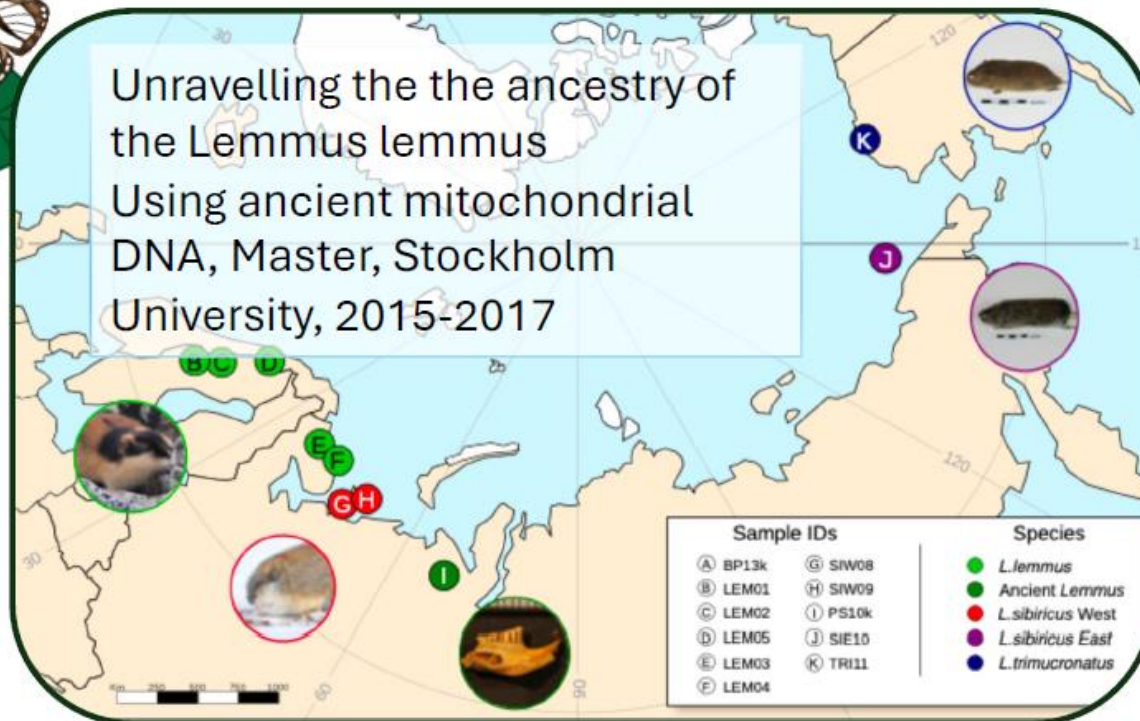
Karin Näsval, PhD

Postdoctoral fellow

Rapid speciation group, Tree of Life
Wellcome Sanger Institute, UK

Unravelling the the ancestry of
the Lemmus lemmus

Using ancient mitochondrial
DNA, Master, Stockholm
University, 2015-2017

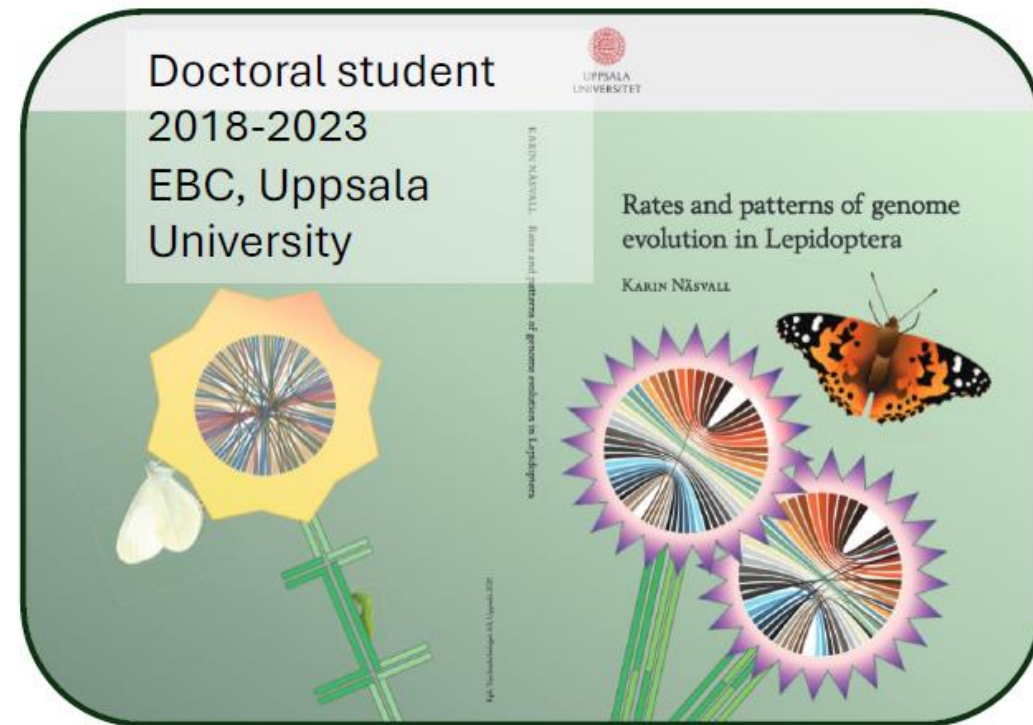


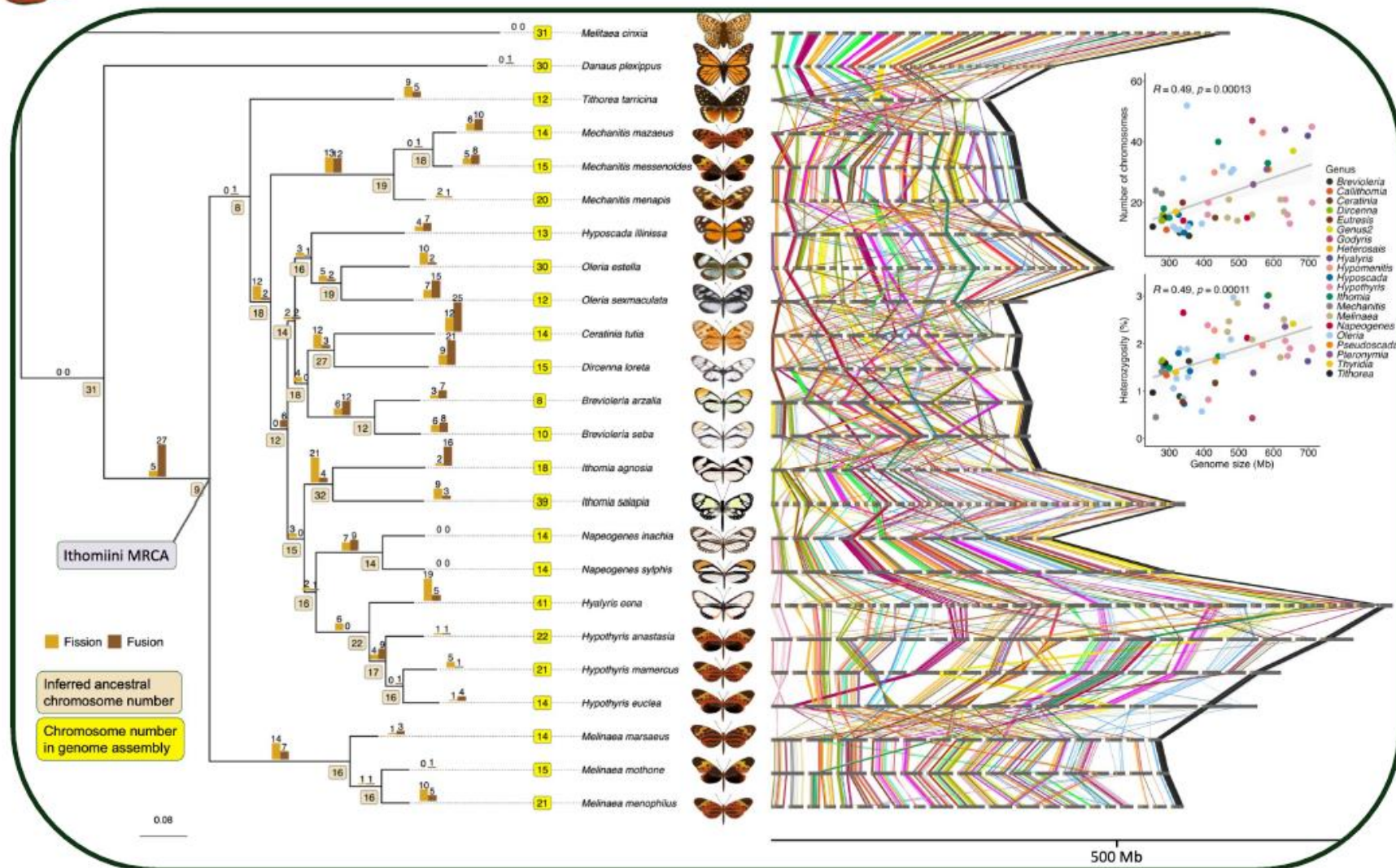
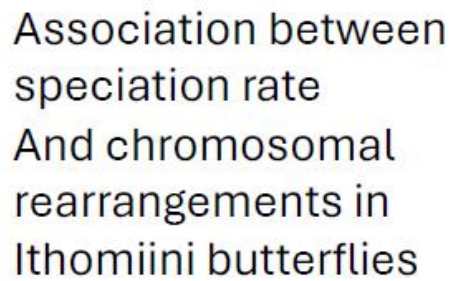
Doctoral student
2018-2023
EBC, Uppsala
University

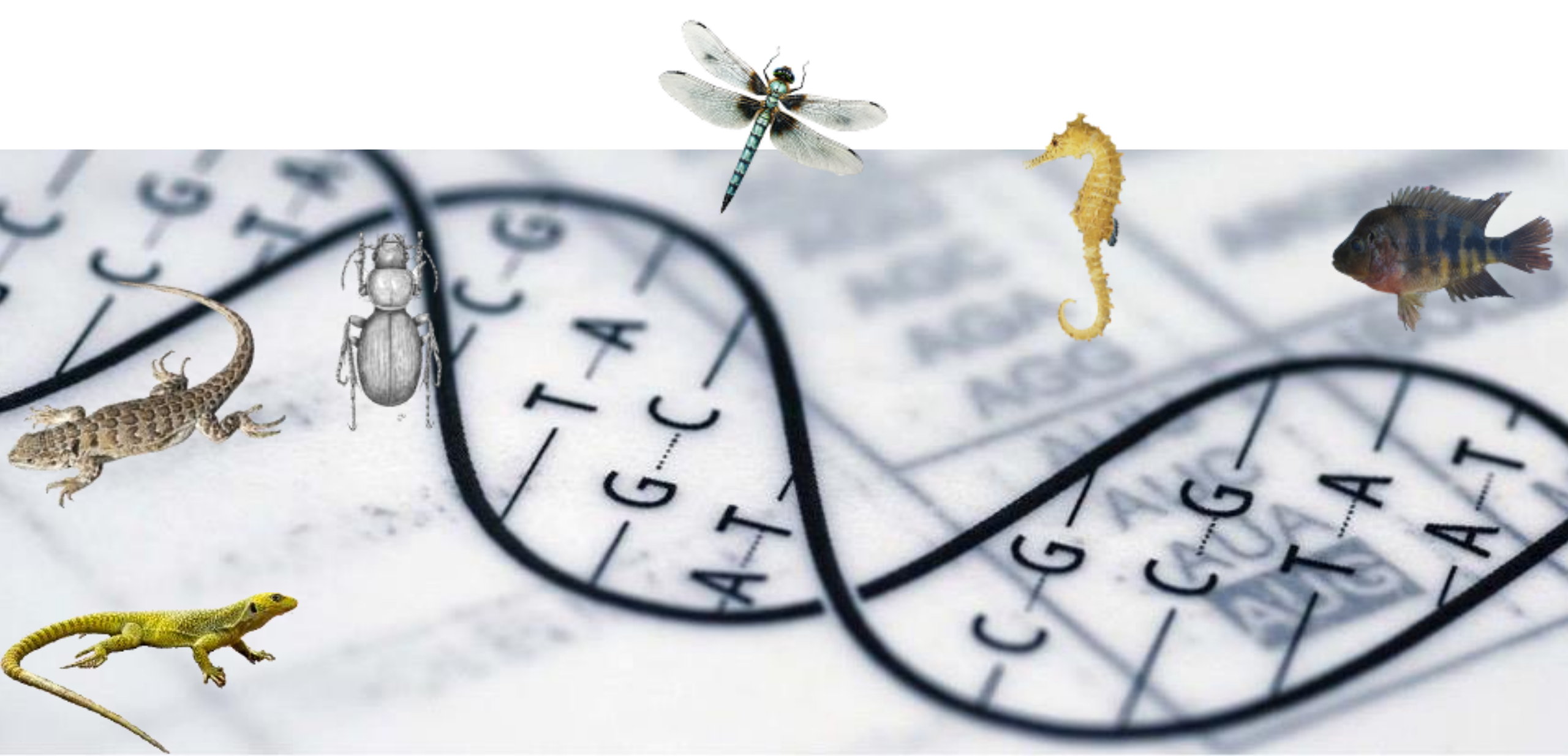


Rates and patterns of genome
evolution in Lepidoptera

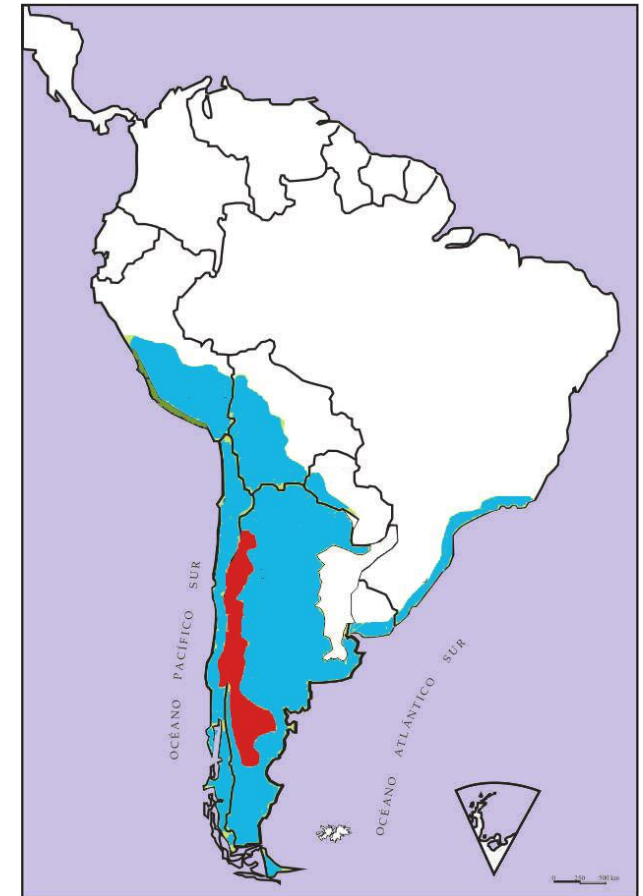
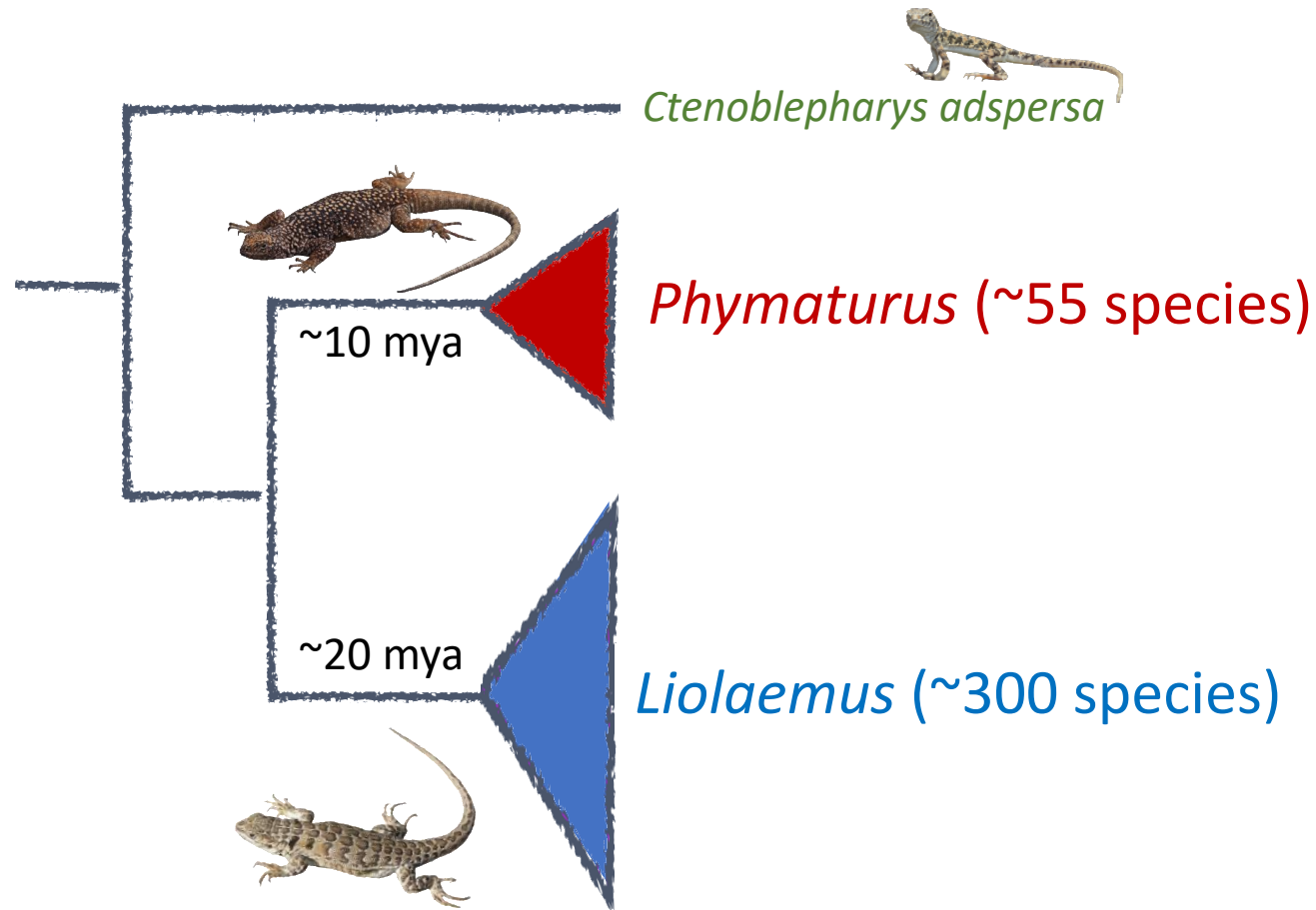
KARIN NÄSVALL







Liolaemus is the temperate-zone genus with the highest number of species among all reptiles, mammals, and birds.





Liolaemus dicktracyi



Dunas Río Paraná



Liolaemus martorii
San Antonio Oeste



Liolaemus tenuis
Chillan



Liolaemus koslowsky
La Rioja



Liolaemus silvanae
Meseta Lago Buenos Aires

Liolaemus (~300 especies)

Photo: Luciano Avila

Introduction to Biodiversity Genomics

Using genomics to understand and preserve biodiversity from genetic diversity, populations, to species and ecosystems

Resolving the taxonomy

- Placing potentially new species
- Species delineation



Adaptation and speciation

- Identifying genomic regions involved in speciation
- Identifying genes underlying traits



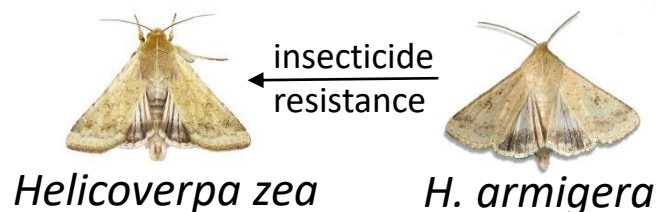
Are the species declining?

- Detecting past inbreeding
- Assessing genetic diversity



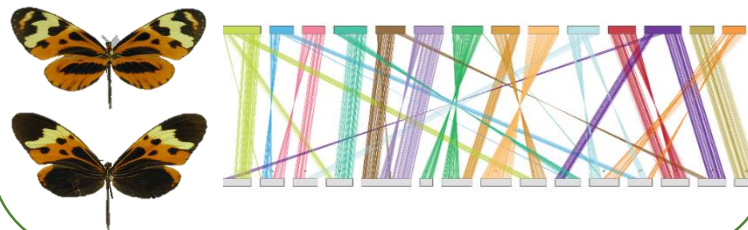
Studying gene flow

- Are populations/species hybridising or have in the past?
- Finding regions of adaptive introgression



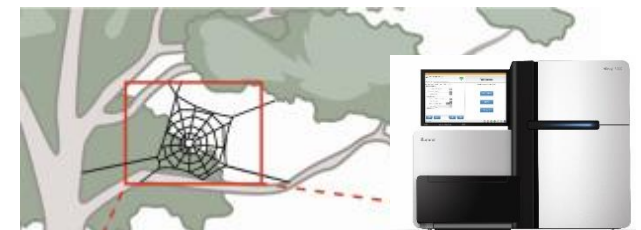
Studying genome evolution

- Gene expansions, e.g. olfactory
- Chromosomal rearrangements
- Genome size evolution (TEs, etc)

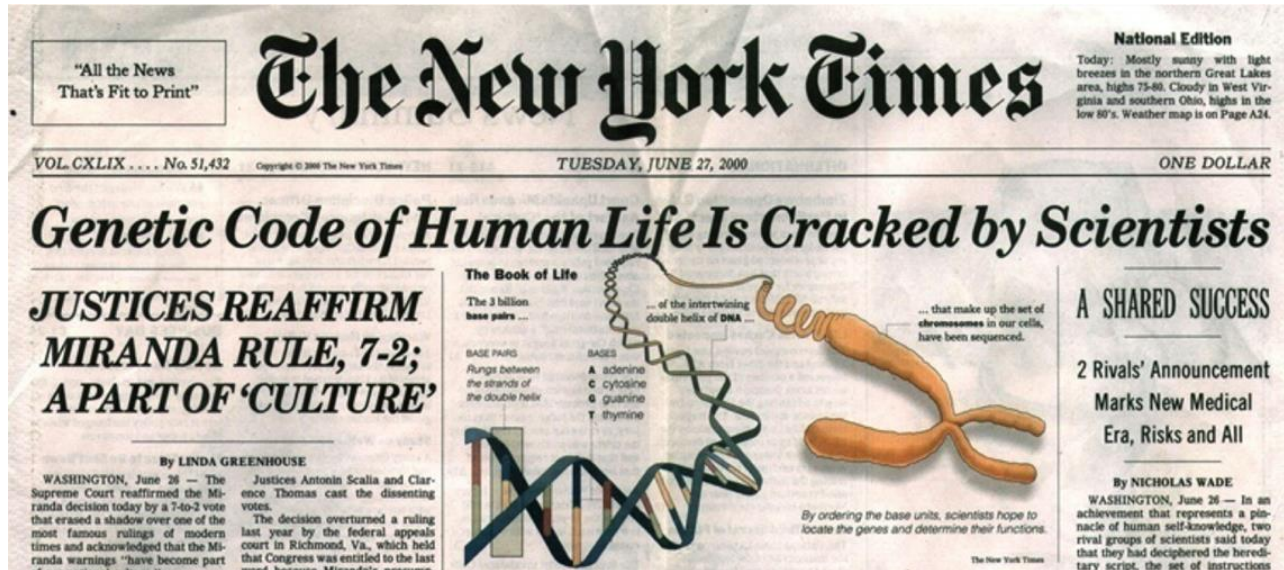


Which species occur here?

- Identifying biodiversity hotspots
- Monitoring effectiveness of conservation strategies



Human Genome Project

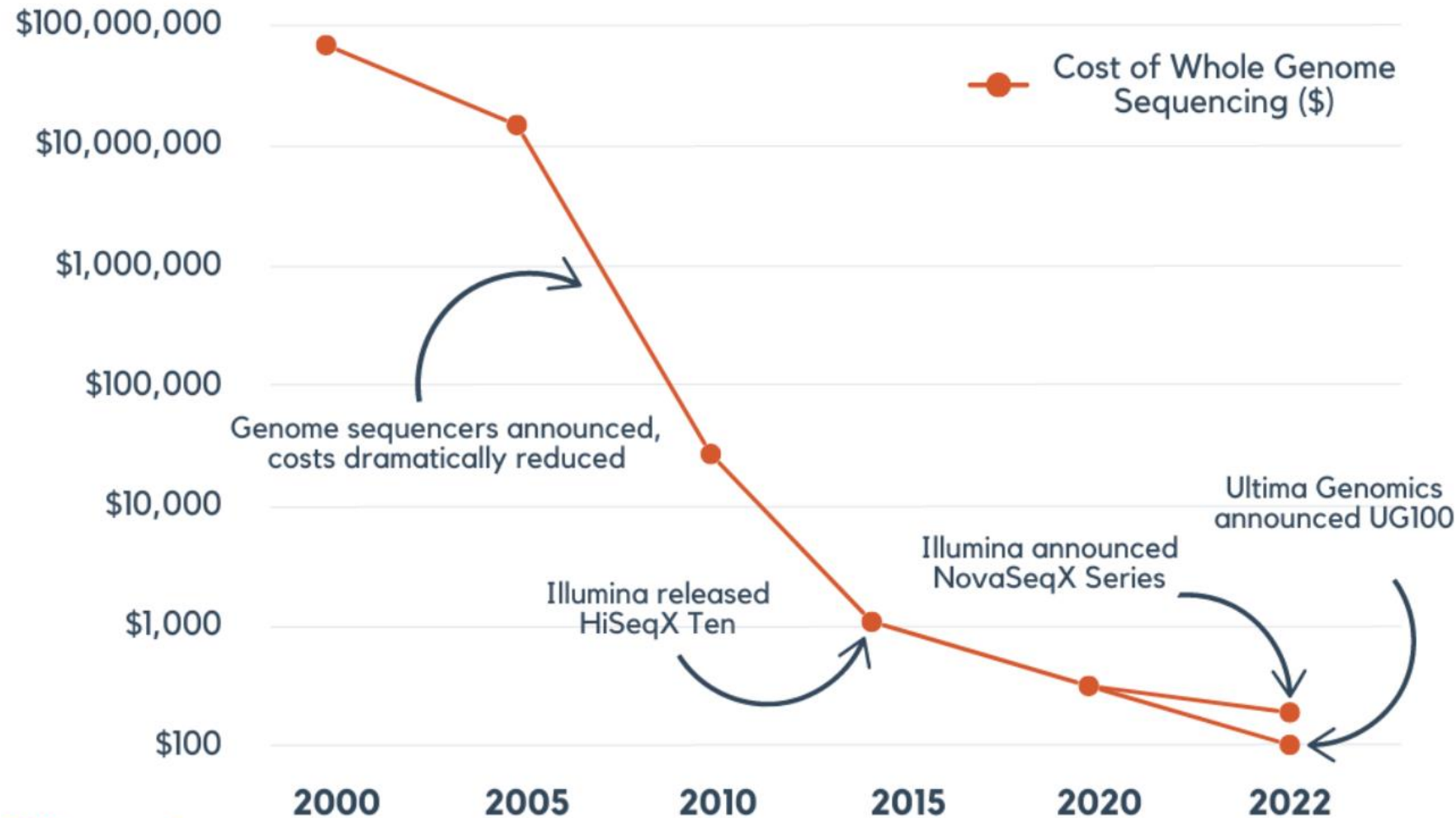


- Human genome project – started in 1990, completed in 2003
- Sequenced across ~20 institutions worldwide
- Cost an approximate \$5 billion US dollars

The first human reference genome transformed modern medicine and understanding of human evolution and physiology

- Comparing populations e.g. to study how humans spread across the globe
- Finding introgression with neanderthals and denisovans
- Identifying genes under selection, like the lactase gene
- Finding genes causing diseases like breast cancer
- Understanding how cancer develops
- Personalised medicine
- Single-cell sequencing to understand which genes are active in which cells

Sequencing costs are decreasing rapidly

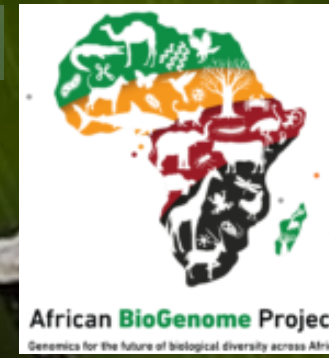


Since Oct 2023
PacBio Revio
(66 Gbp per lane
in 15 kb reads)





Project Psyche



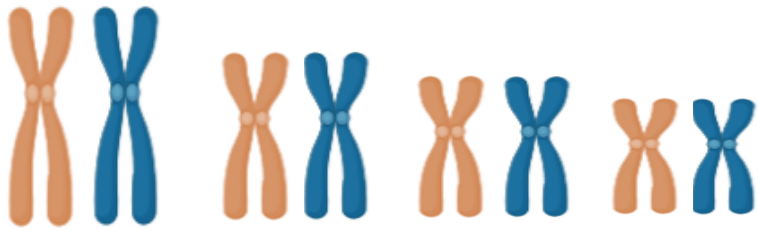
CREATING A NEW FOUNDATION FOR BIOLOGY

Sequencing Life for the Future of Life

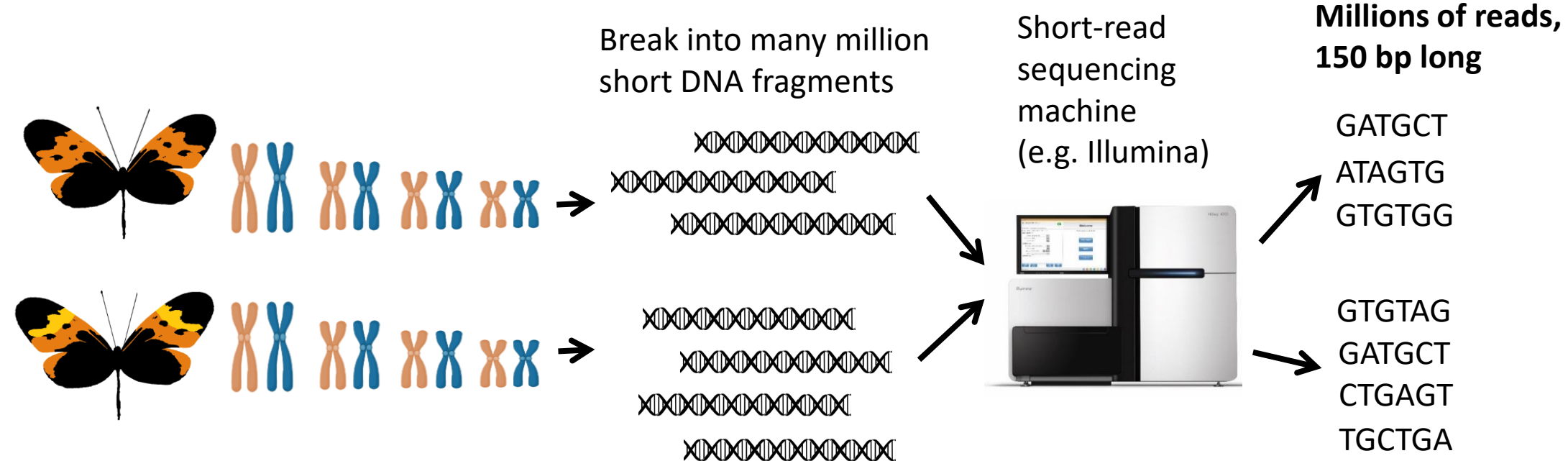


Why do we need a reference genome for whole-genome sequencing projects?

Genome = set of all chromosomes

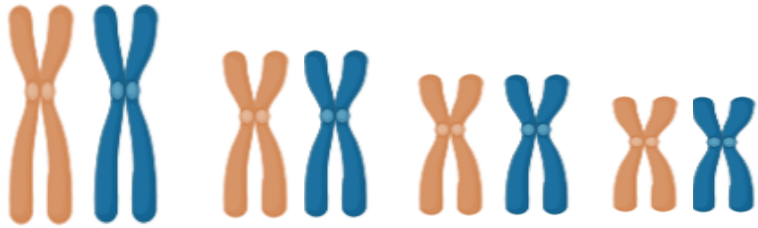


Problem:
**We do not know which
of these sequences to
compare**



How do we make a reference genome?

Genome = set of all chromosomes



Break into many million
long (10-50 kbp) DNA fragments



Long-read sequencing
machine (e.g. PacBio)

more expensive than
short-read sequencing



**Millions of reads,
10-20 kbp long**

GATGCTGAGTA
ATAGTGTGGAT
GTGTGGATGTG
TGCTGAGTTCG
TGGATGCTGAT
CTGAGTTCTCG

Reference genome

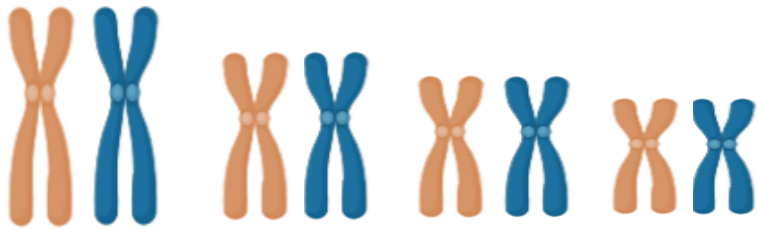
ATAGTGTGGATGCTGAGTTCGT

ATAGTGTGGATG
GTGTGGATGCT
TGGATGCTGAGT
GATGCTGAGTTC
TGCTGAGTTCG
CTGAGTTCGT

puzzling them together
(aligning them to
each other)

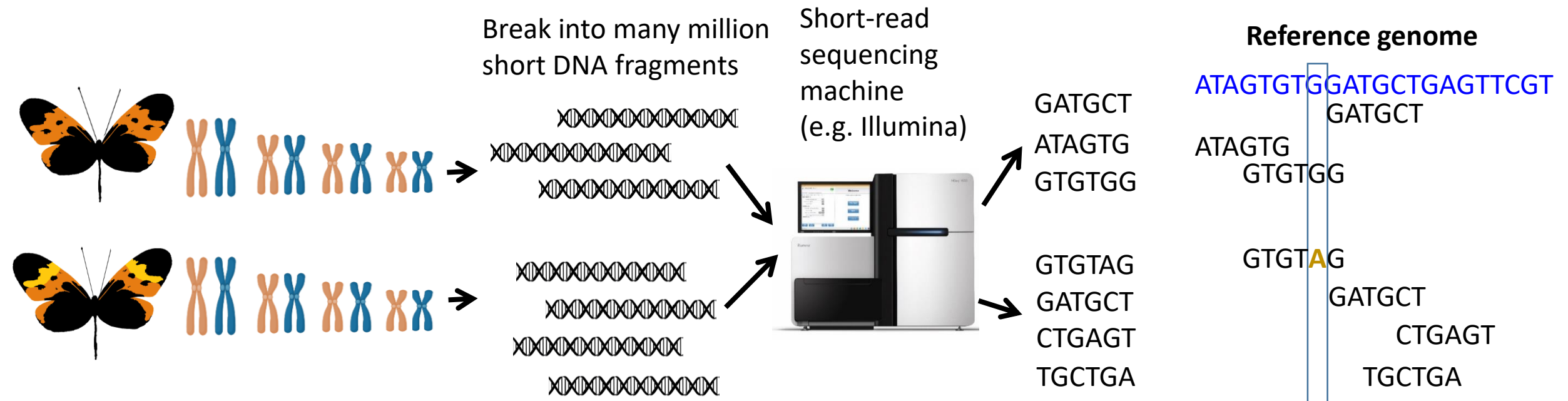
How do we use this reference genome?

Genome = set of all chromosomes



Solution:

The reference genome allows us to place the reads so that we can compare them across individuals, populations or species

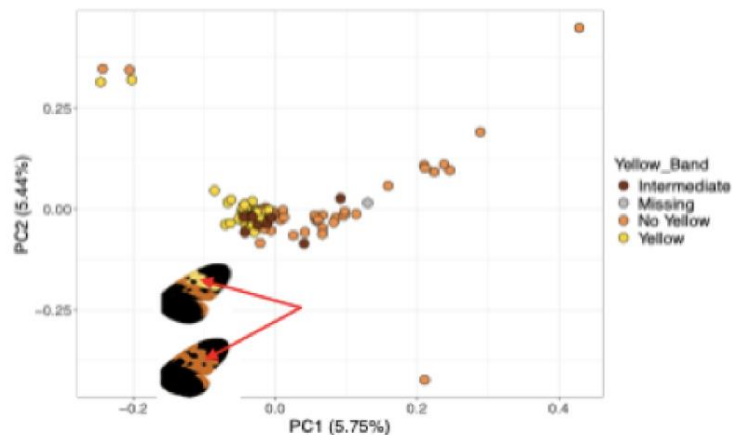




Do these two butterflies belong to different species?



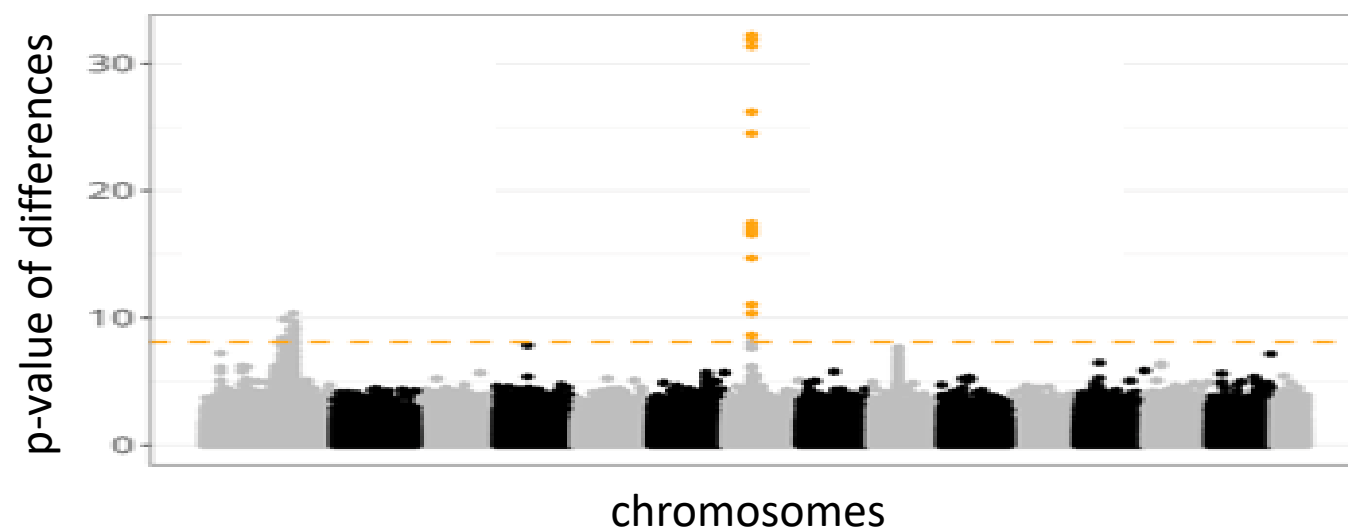
Eva van der Heijden



CRISPR butterflies with *ivory* knocked-out

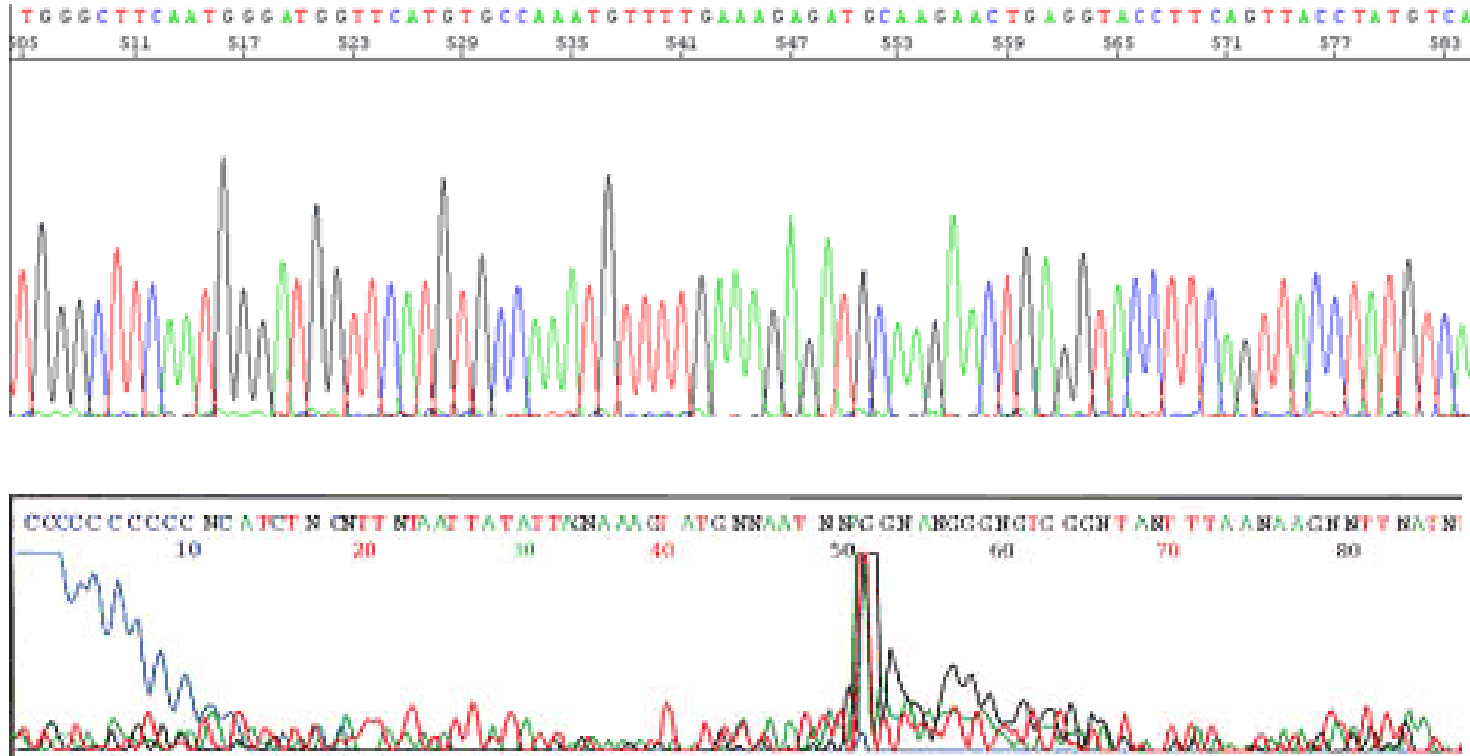


They differ only at one region in the genome



Introduction to high-throughput or next-generation sequencing

Sanger Sequencing (since 1980s)



- Possible to manually check each sequence and resequence failed sequences
- Requires primer sequences and has very low throughput (expensive per bp)

Two main types of high-throughput sequencing

- **Short-read sequencing**

- Reads are typically 150 bp long
- Cheaper than long-read sequencing
- E.g. Illumina, soon probably also Ultima Genomics

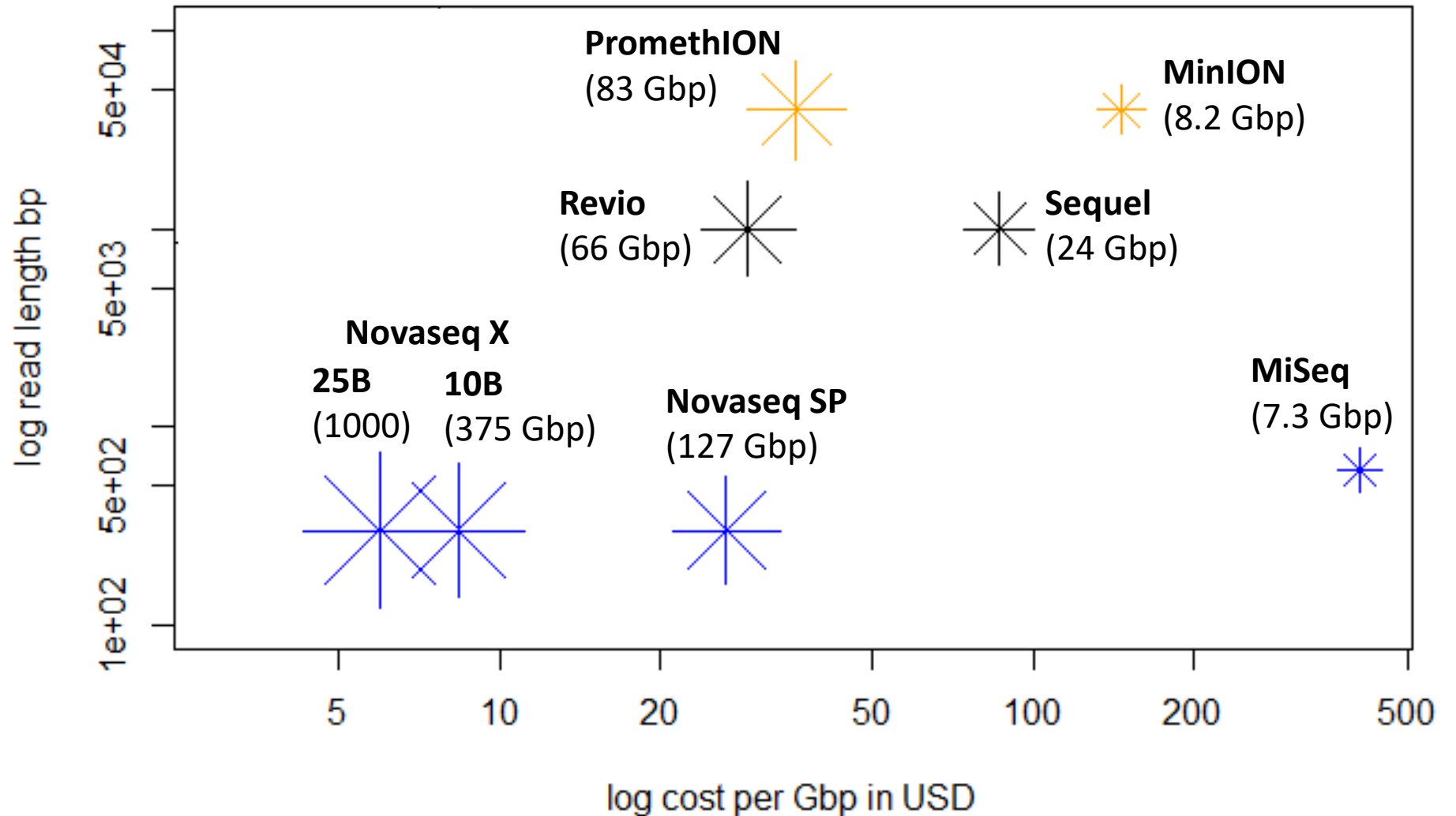
- **Long-read sequencing**

- Reads are typically >10 kb long (PacBio: 15-20 kbp, Nanopore: 10-100 kbp)
- More expensive than short-read technologies
- Required for making a reference genome
- E.g. PacBio or Nanopore

Read length versus per Gbp sequencing costs for different sequencing machines (note the axes are in logarithmic scale)

* ONT
* PacBio
* Illumina

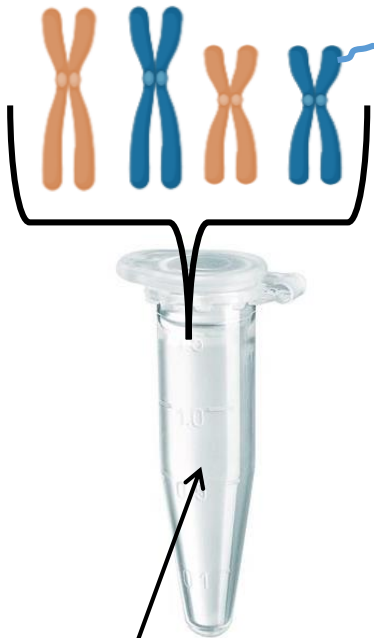
Star size shows the total throughput per lane, also given in parentheses ()



Whole-genome sequencing

Genome

= complete set of chromosomes

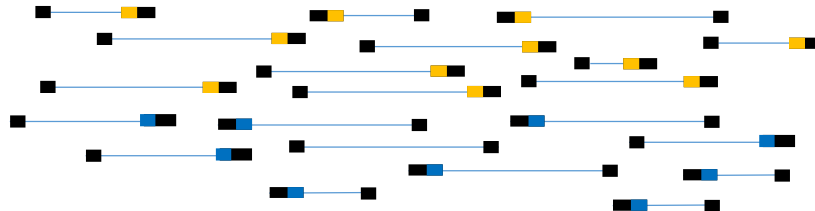


DNA (chromosomes)

Breaking chromosomes into shorter pieces for sequencing



Adding sequencing adapters incl. individual index



Size selection



Long-read sequencing (PacBio or Nanopore/ONT)
paired-end sequencing



10-50 kbp



Short-read sequencing (e.g. Illumina)
paired-end sequencing



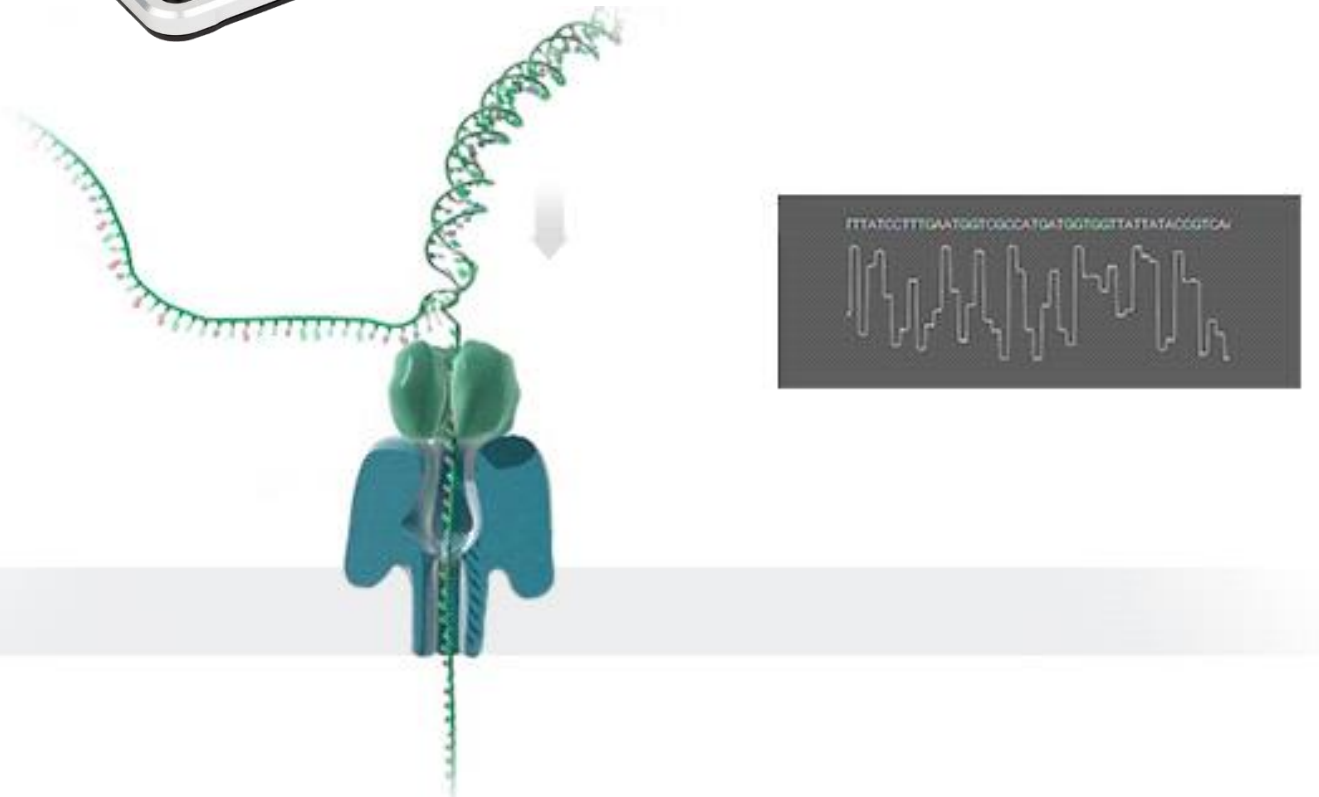
350-500 bp



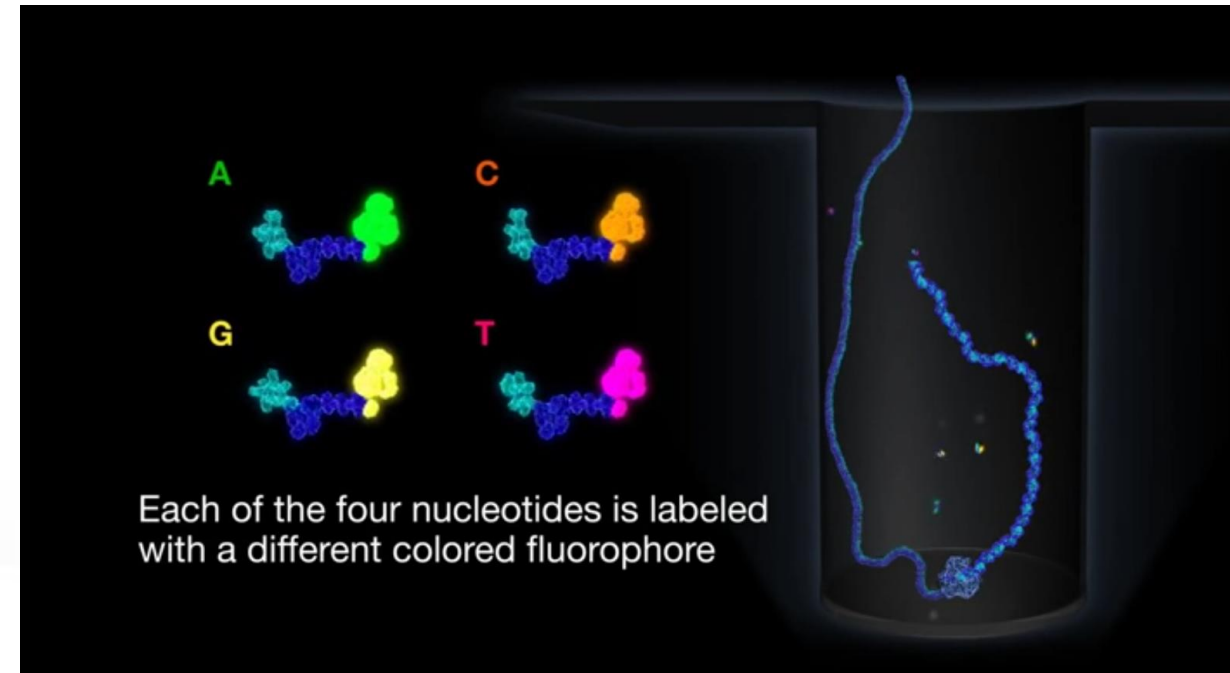
Long read sequencing technologies



Nanopore



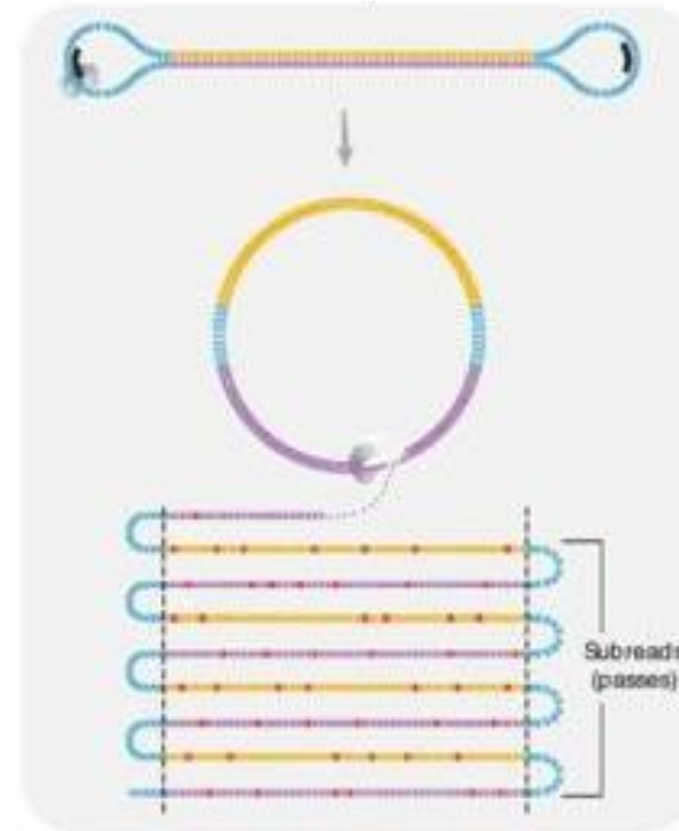
PacBio



PacBio HiFi reads (99.95% accuracy)

Each DNA-fragment is sequenced many times to get a high-quality consensus (=summary) read

Multi-pass sequencing
on Sequel II System



HiFi Read Base Calling

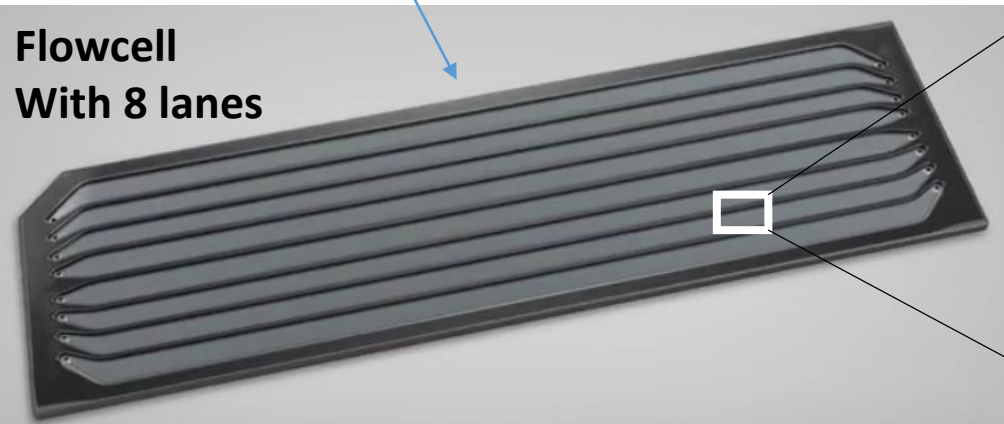
HiFi READ
Read Lengths up to 25,000 bp
Average Read Accuracy $\geq 99.5\%$



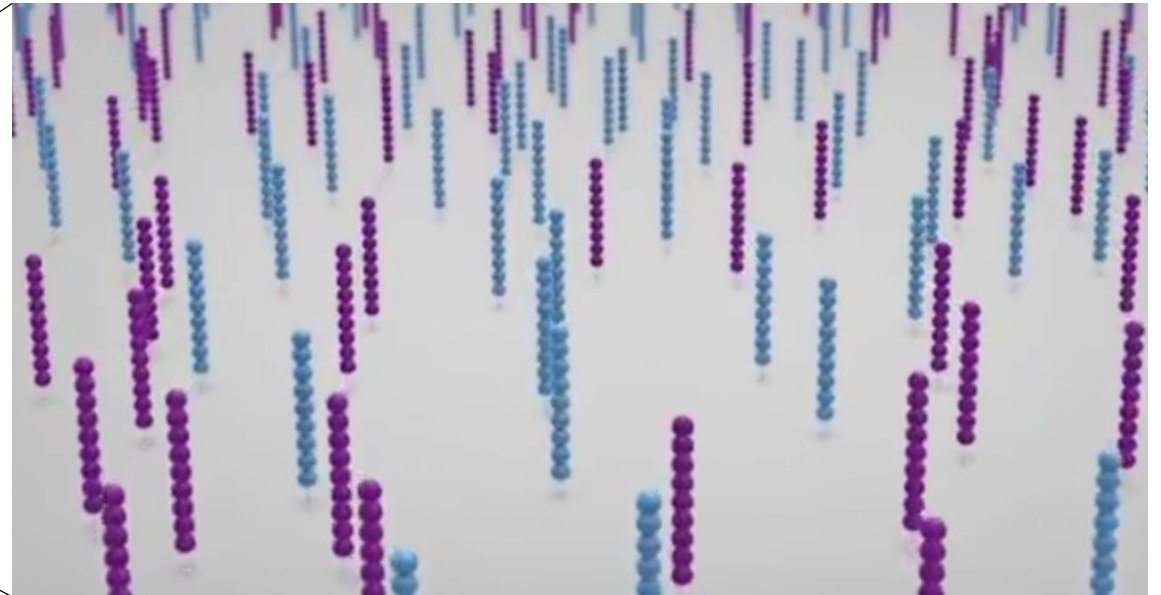
Illumina flowcell

DNA fragments with Illumina adapters

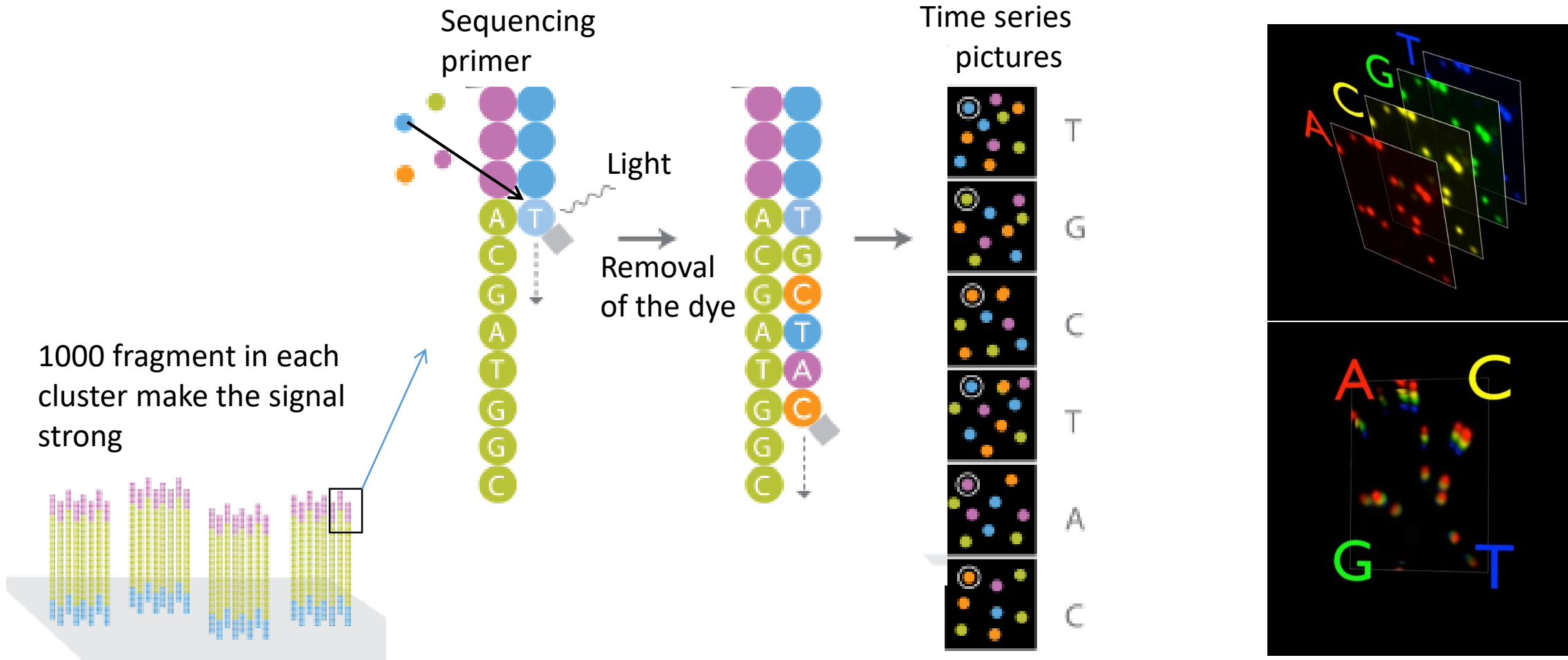
Flowcell
With 8 lanes



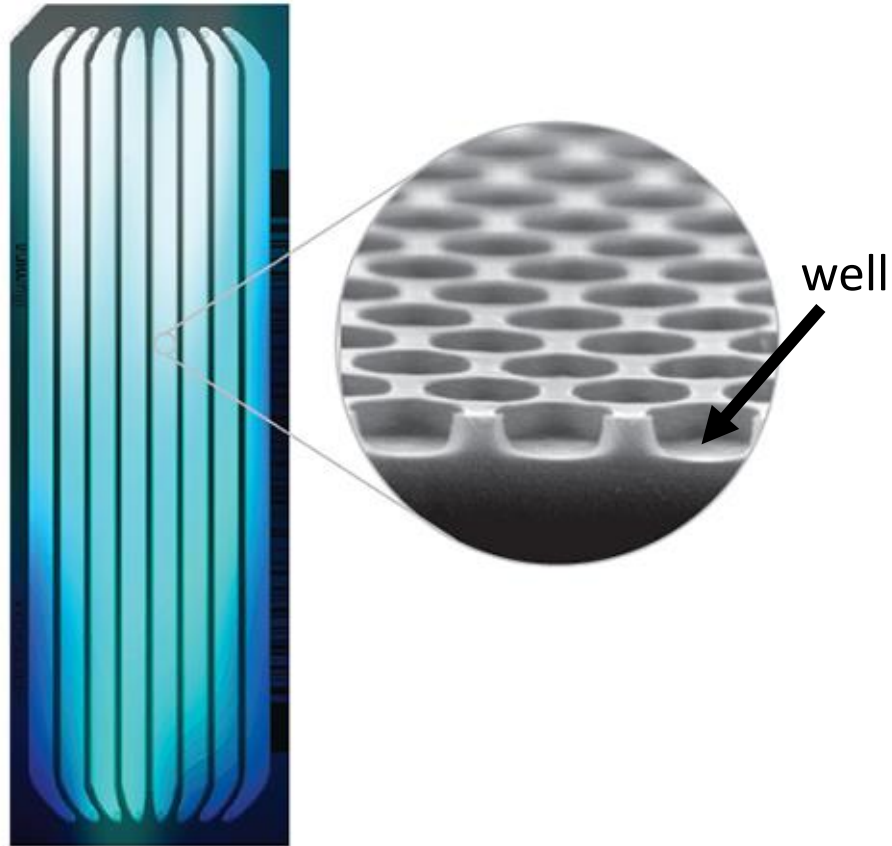
Each lane contains a dense lawn of Illumina primers



Short-read sequencing with Illumina



Newer Illumina machines use wells and only 2 colours (e.g. Novaseq, Nextseq, MiniSeq. This makes it faster and cheaper)

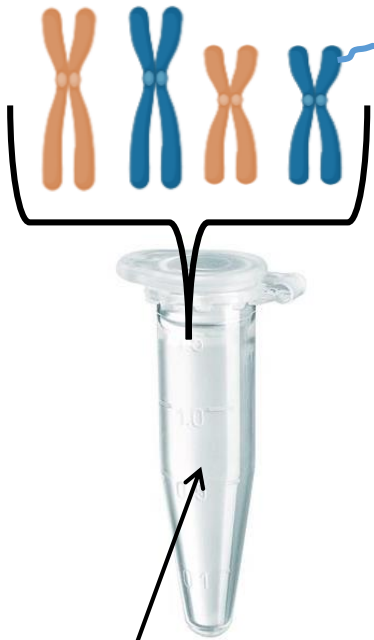


2-Channel Chemistry				
	 A	G	 T	 C
Image 1				
Image 2				
Result	A	G	T	C

Whole-genome sequencing

Genome

= complete set of chromosomes

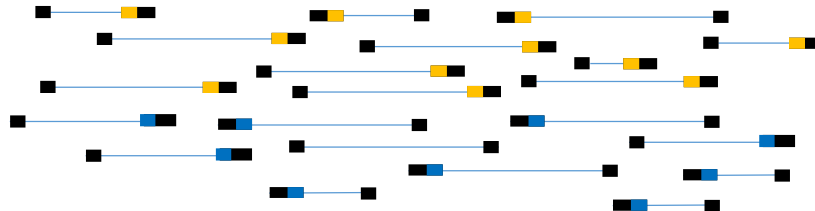


DNA (chromosomes)

Breaking chromosomes into shorter pieces for sequencing



Adding sequencing adapters incl. individual index



Size selection



Long-read sequencing (PacBio or Nanopore/ONT)
paired-end sequencing



10-50 kbp



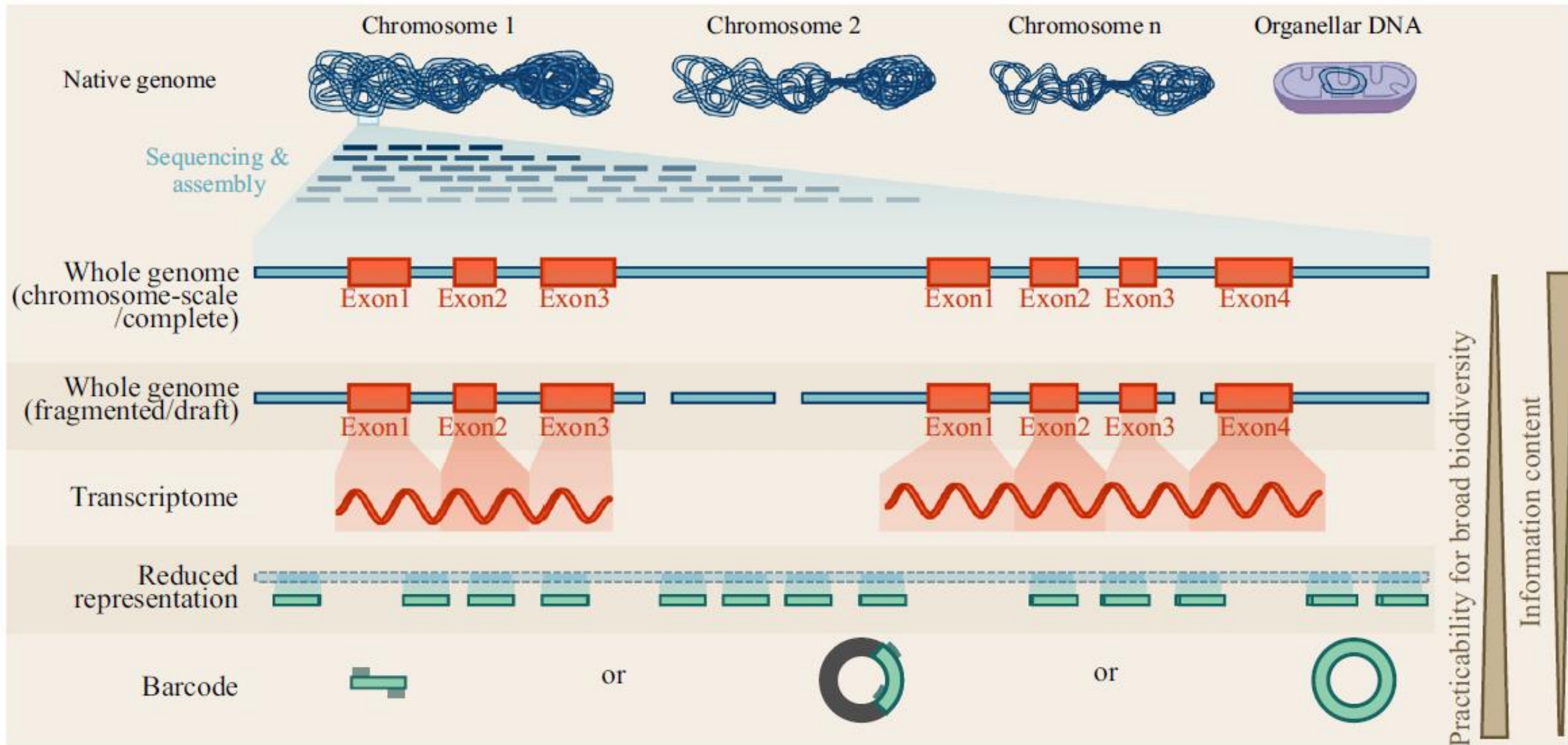
Short-read sequencing (e.g. Illumina)
paired-end sequencing



350-500 bp



Sequencing approaches for biodiversity genomics

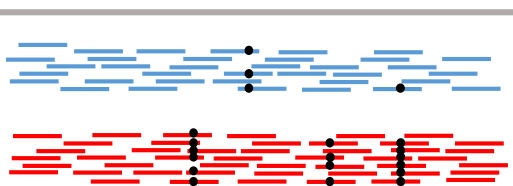


Sequencing approaches for biodiversity genomics

Whole-genome resequencing (short-read data)

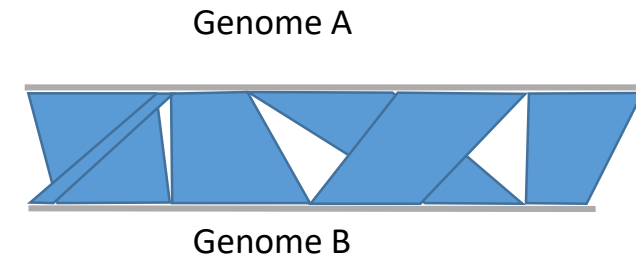
- Requires a reference genome
- individuals need to be from the same or closely related species
- Complete genome sequenced

reference
genome



Genome assembly comparisons (long-read data)

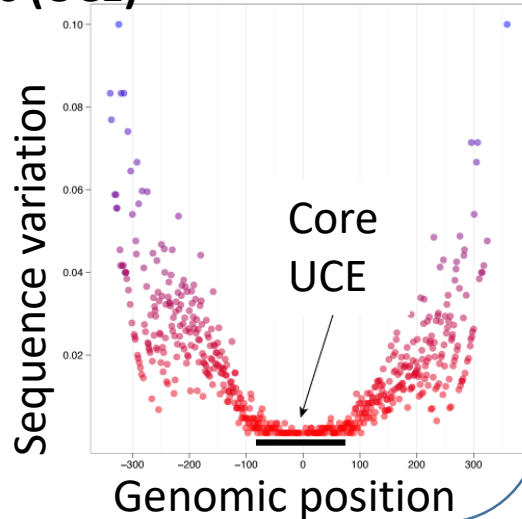
- Comparative genomics – studying structural variation between species, can be distantly related
- Gene expansions, transposable elements etc
- Phylogenomics across deeply divergent species
- Pangenomics – multi-genome assemblies to study within-species variation in structural variants



Reduced-representation techniques (only parts of the genome sequenced)

Ultra-conserved elements (UCE)

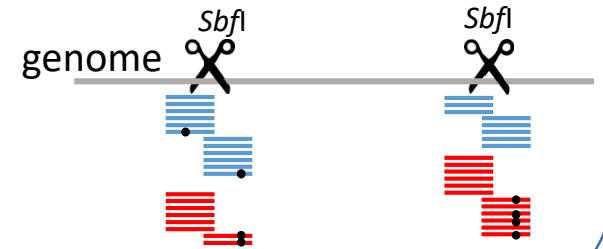
- Sequence capture with baits based on genomic regions that are conserved across many species
- Works with highly divergent species



Restriction Associated Sequencing (RAD)

(similar methods: GBS, ddRAD)

- does not require primers/baits or reference genome
- individuals need to be from the same or closely related species
- Information from thousands of loci distributed across the genome

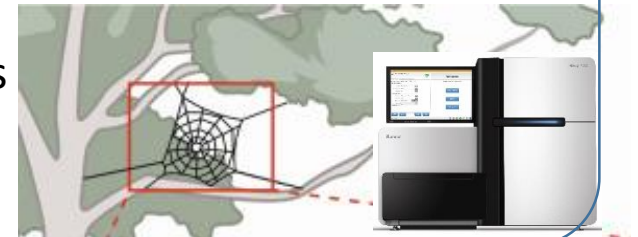


Targeted or amplicon sequencing, e.g. barcoding

- Sequencing one or few genes
- requires primers
- e.g. CO1 (mitochondrial barcoding region), advantage: large database (BOLD) available to compare to for species identification

Environmental DNA (eDNA)

- Mostly CO1 sequencing from soil, water, air (spider webs)
- Identifying local species
- Studying species richness



Trade-offs: Splitting reads (i.e. costs) among:

Total data gets divided by:

- Number of sites to sequence
 - Depends on genome size and sequencing strategy, e.g. RAD versus whole-genome
- Sequencing depth (e.g. sequencing at 10x depth of coverage)
- Number of specimens to sequence
- Example: 1 Novaseq X 10 B lane
~2.5 billion paired-end reads of 150 bp each -> 375 Gbp data
 - 100 whole-genomes of a species with 0.375 Gbp genome size at 10x coverage
 - 19 whole-genomes of a species with 1 Gbp genome size at 20x coverage
 - 375 individuals sequenced with a RAD sequencing approach resulting in 50 Mbp at a sequencing depth of 20x

**Now let's get started with
handling genomic data!**

**First, a brief overview of what
we will do now**

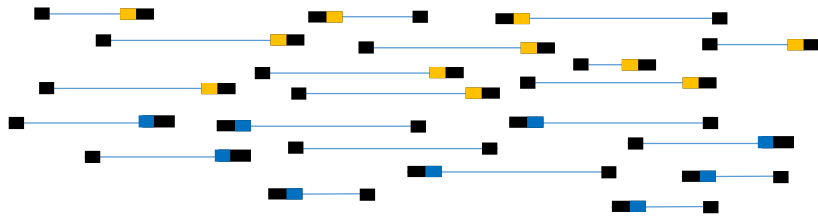
Whole-genome short-read sequencing

DNA

Random shearing



Illumina adapter ligation
incl. individual index



Size selection



paired-end sequencing

Up to 20 billion read pairs

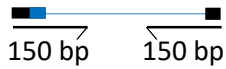


1. Quality check and trimming raw reads

paired-end Illumina sequencing



Up to 20 billion
read pairs



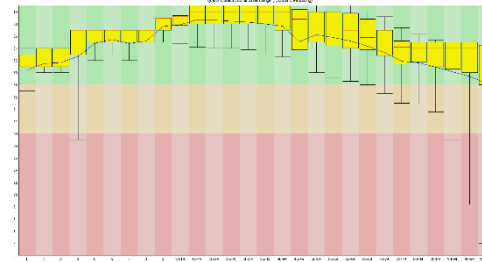
Forward
reads



Reverse
reads

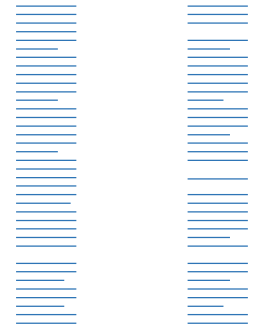


Fastqc quality plots



fastp

Trimmed and filtered
reads of good quality



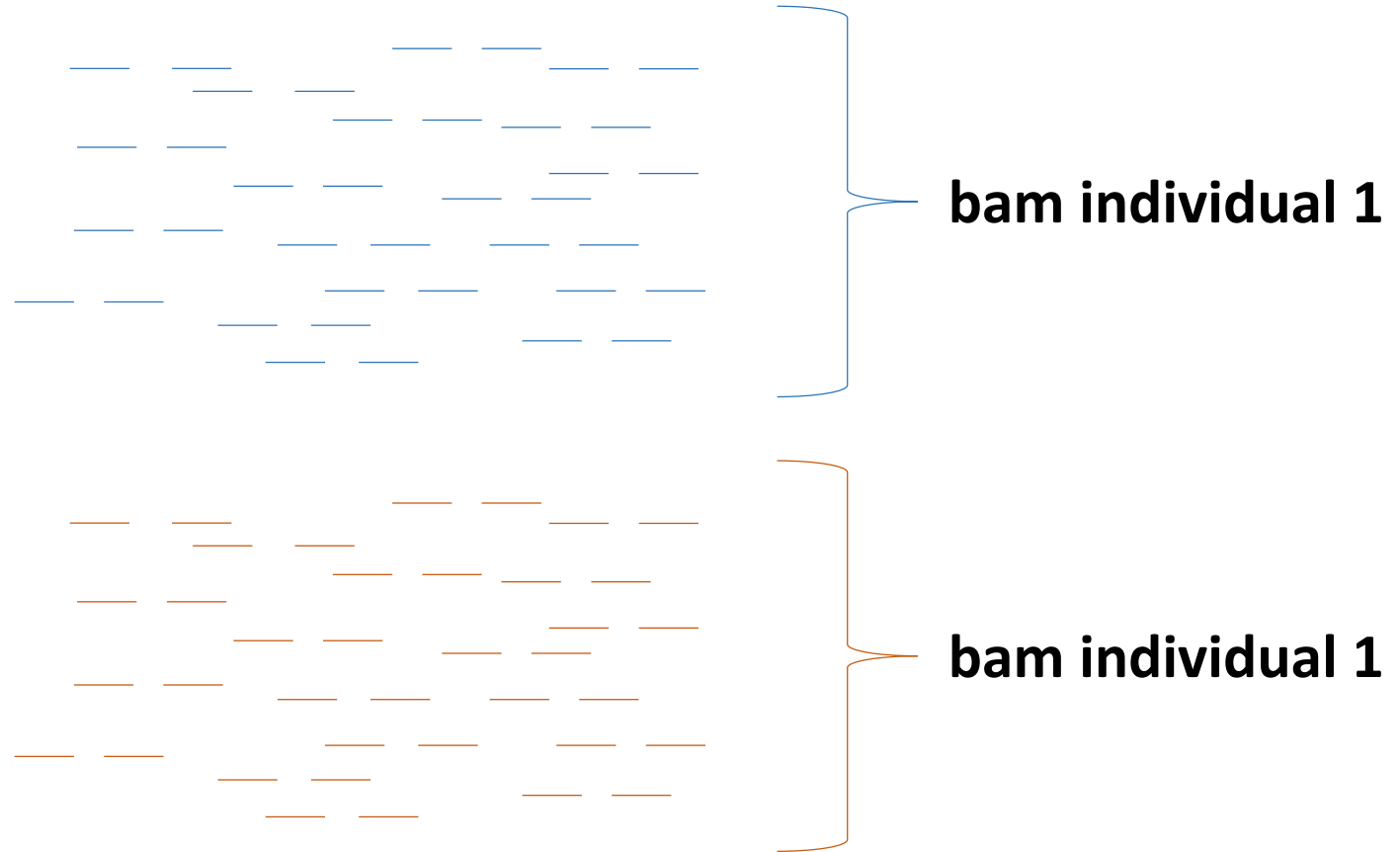
Fastq file of reads

```
@HWUSI-EAS611:34:6669YAAXX:1:1:5069:1159 1:N:0:  
TCGATAATACCGTTTTTTTCCGTTTGATGTTGATACCAT  
+  
IIHHIIHIIIIIIIIIIIIIIIIIIIIIIIIHIIIIHIIIII
```

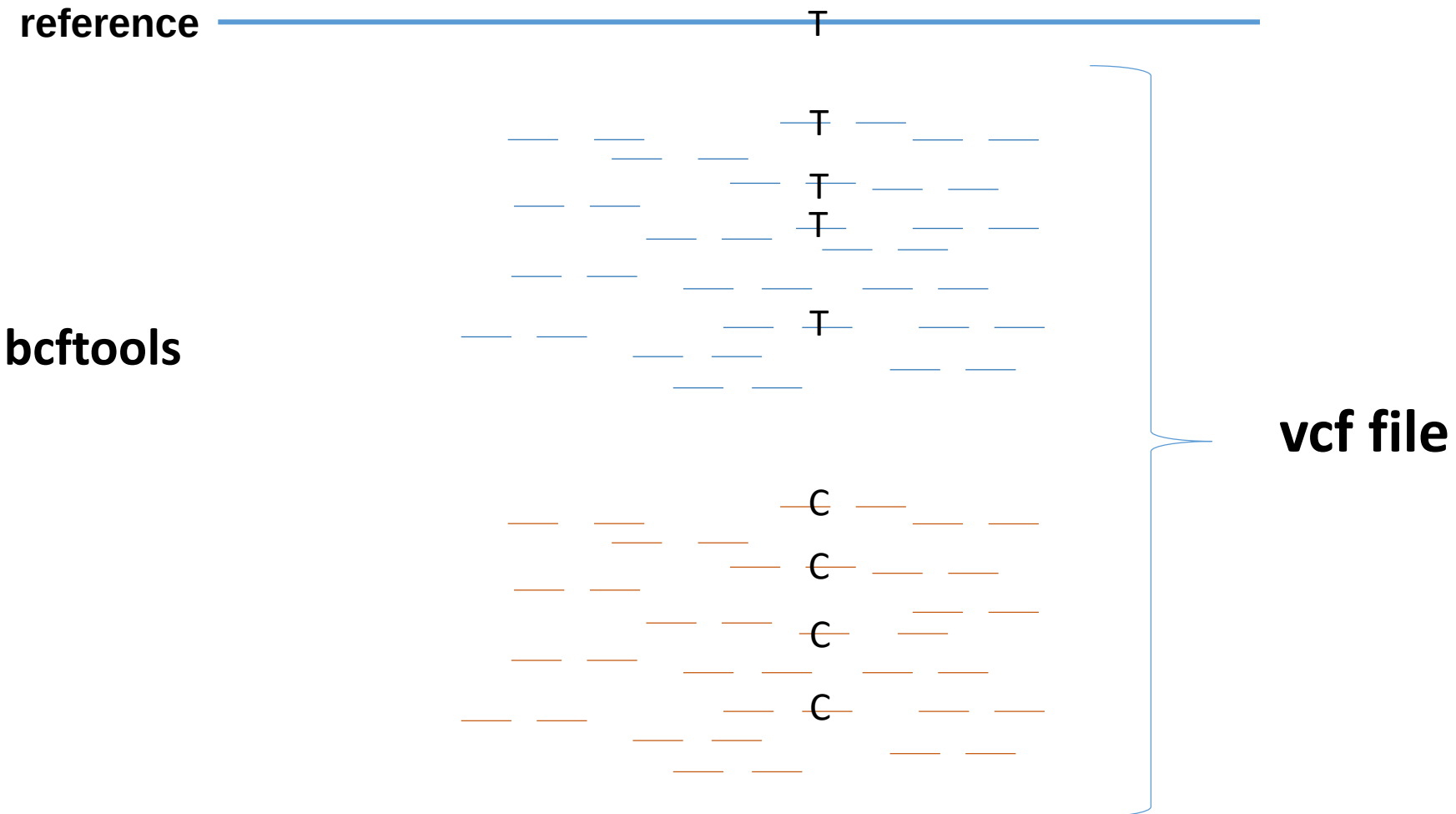
2. Alignment to the reference genome with bwa

reference

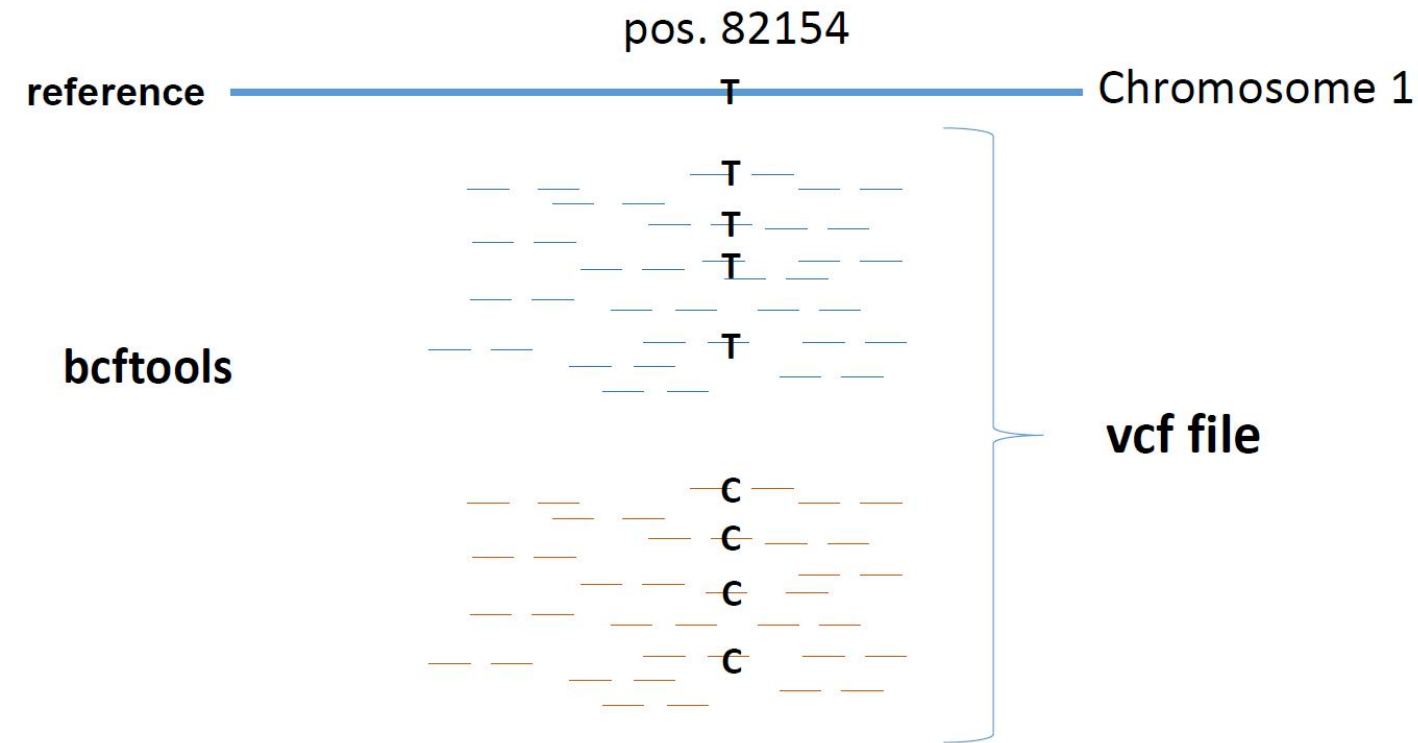
bwa



3. Variant and genotype calling with bcftools



3. Variant and genotype calling with bcftools



vcf file: Genotypes for each individual at genomic sites

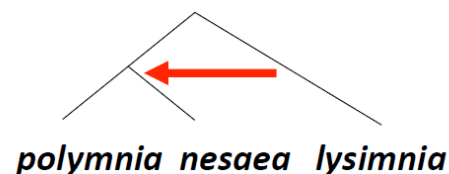
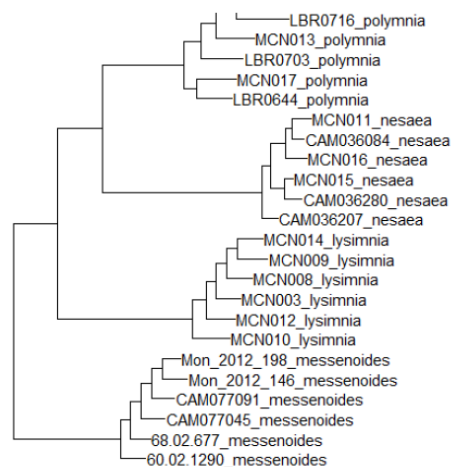
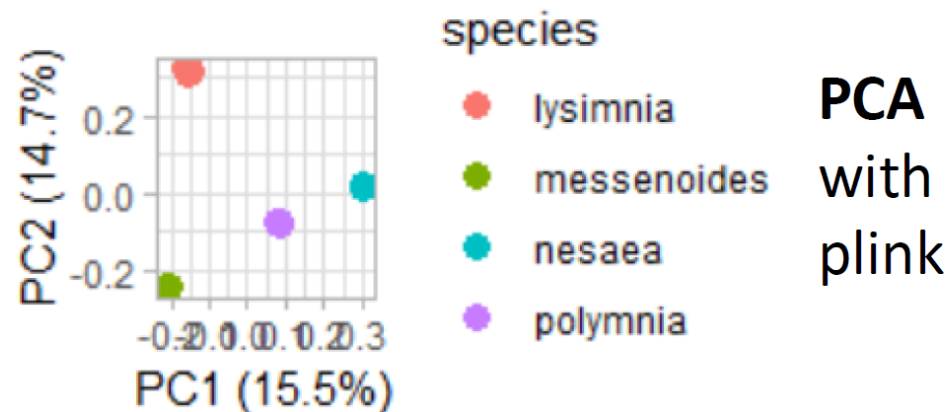
```
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT GEN
chr1 82154 . T C . GT 0/0 1/1 0/0 1/1 0/0 1/1
chr1 752566 . T . . GT 0/0 0/0 0/0 0/0 0/0 0/0
chr1 752721 . T C . GT 1/1 1/1 1/1 1/1 1/1 1/1
chr1 752721 . A . . GT ./.
```

Using the vcf file for biodiversity genomics

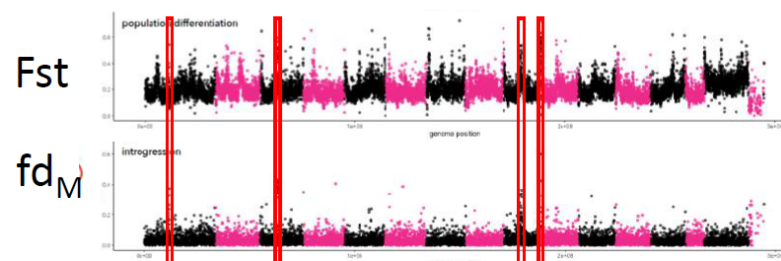
vcftools:
Filtering sites and genotypes

Filtered vcf

Chr Pos R A ind1 ind2 ind3
Chr1 1 A T 0/0 0/1 1/1
Chr1 3 A . 0/0 0/0 ./.



Testing for introgression
with Dsuite (ABBA-BABA test)

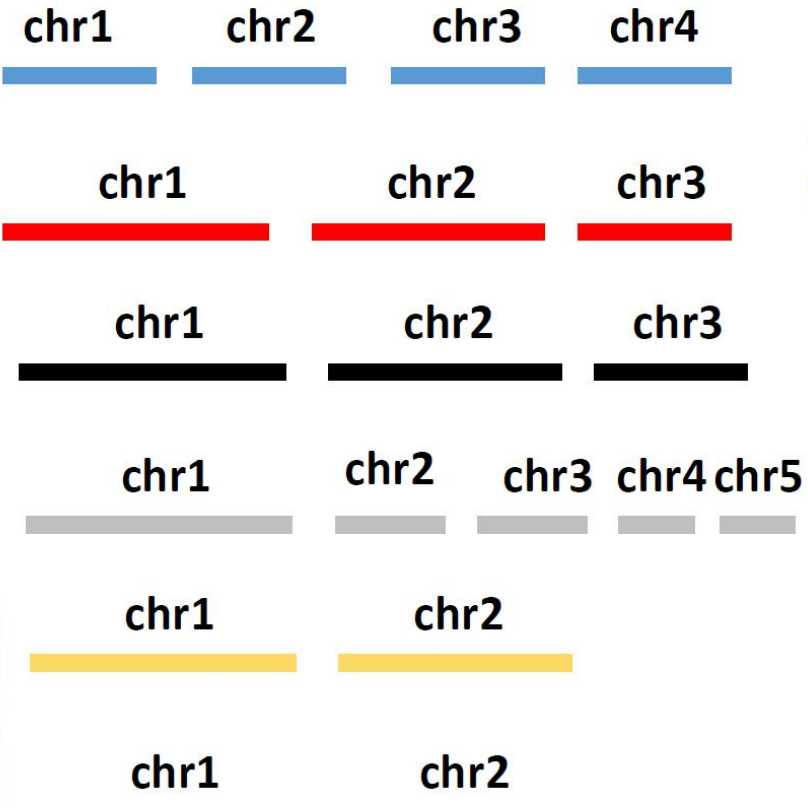


Genome scans for differentiation (F_{ST} , D_{XY}) and introgression (f_d)

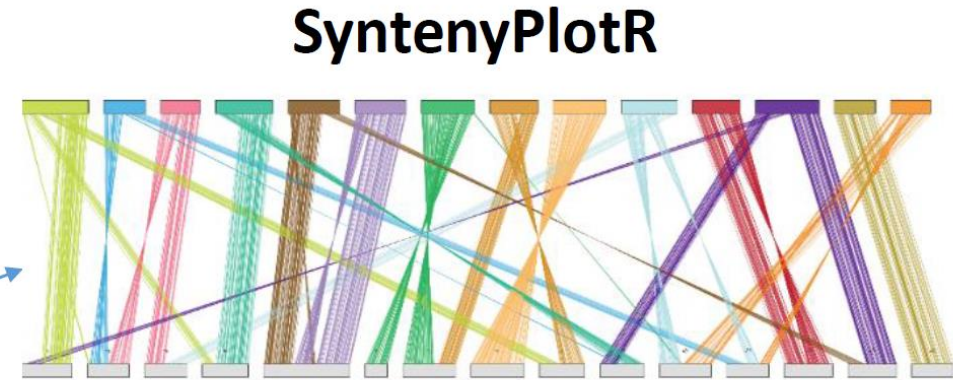
Comparative genomics

Download genomes from
<https://www.ncbi.nlm.nih.gov/>

Fasta file (.fna)



Minimap2:
whole-genome
alignment



Orthofinder + Mafft

Fasta alignment file
per BUSCO gene

gene1	gene2	gene3
>ind1	>ind1	>ind1
ATCTG	CTGAA	CCCTA
>ind2	>ind2	>ind2
ATCTC	CTGGA	CCGTA
>ind3	>ind3	>ind3

Gene trees with
iqtree

